

ASSIGNMENT 3: Neural Network Classifiers**(Group consists of 8 students)****Prof. Varun Dutt****Due: Before 11:59 PM on August 28, 2026****Readings:**

- Class notes and slides from week 4, 5 and 6
- Witten, Frank, and Hall (2011):
[https://www.dropbox.com/s/j5jxyfiw1nsv5xy/%5Blan H. Witten%2C Eibe Frank %2C Mark A. Hall%5D Data Min%28BookFi.org%29.pdf?dl=0](https://www.dropbox.com/s/j5jxyfiw1nsv5xy/%5Blan%20H.%20Witten%20Eibe%20Frank%20Mark%20A.%20Hall%5D%20Data%20Mining%28BookFi.org%29.pdf?dl=0)
- Weka website: <http://www.cs.waikato.ac.nz/ml/weka/>

Objectives:

- Classification of datasets using Neural Network Classifiers.
- Try Neural Network classifications using WEKA.

Assignment:

1. Please see dataset containing the following training patterns: [12]

Training Pattern	Attributes		Desired Output
	Input 1	Input 2	
1	1	1	1
2	0	1	1
3	1	0	1
4	0	0	0

Now, assume the following definition of output

$$y = \begin{cases} 0 & \text{if } I < 0.5 \\ 1 & \text{otherwise} \end{cases}$$

Where, I is the overall input at a node in the network.

Please find out whether classes represented in the desired output are linearly separable and can be modeled by a neural network with one node in a single layer. Please show your working.

Now, propose a suitable neural network for the dataset above. Please assume a random value of weights (ideally between 0 and 1) in the network and a learning rate (β) = 0.2 in the neural network. Using case-based updating, please show the change in weights in each epoch as well as the final set of weights for the neural network.

Finally, please use the final set of weights and the output function to derive predictions from the neural network for the training dataset shown above. Does the neural network classify all the training patterns correctly?

Now, derive predictions for the dataset shown below. How many of the patterns below does the neural network correctly classify?

Training Pattern	Attributes		Desired Output
	Input 1	Input 2	
1	1	1	0
2	0	1	1
3	1	0	1
4	0	0	0

- Download WEKA (if not already done so) and first please read pages 469-472 from Witten, Frank, and Hall (2011) [15]

Also, please watch the YouTube video:

<https://www.youtube.com/watch?v=mo2dqHbLpQo>

Now, please see the training patterns shown below.

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	Clogs	Student
T6	Quiet	Bakcd	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

Upload the training dataset above into WEKA. Now, among classifiers present in the functions group, choose multilayer perceptron. Once the perceptron is loaded, you may click the perceptron name and enable GUI to be true from the menu. Now, please apply the multilayer algorithm to the training dataset and put the initial picture of the resulting neural network. How many input nodes, hidden layers and hidden nodes, and output nodes does this neural network have? What is the default value of the Learning Rate and Momentum value? What is the default value of maximum epochs?

- A. Please execute the default neural network and report the resulting error per epoch. How accurate is the neural network (i.e., how many of the training patterns does the neural network correctly classify)?
- B. Now, do another round of neural network run, where you may change the Learning Rate and Momentum value to be 0.8 respectively. Again, please report the accuracy of the resulting neural network as well as the error per epoch.
- C. Next, to the neural network in A. please add two more nodes in the hidden layer and run the neural network again for the following values of learning rate and momentum parameters: 0.2 each and 0.8 each. Again, please report the accuracy of the resulting neural network as well as the error per epoch for the different values of learning rate and momentum parameters.
- D. Next, to the neural network in C. please add another hidden layer with two nodes and run the neural network again for low (0.2) and high (0.8) values of learning rate and momentum parameters. Again, please report the accuracy of the resulting neural network as well as the error per epoch for the different values of learning rate and momentum parameters.

Finally, please apply the three trained neural networks (in A, C and D) above to the test dataset given below and indicate whether the neural network classify them correctly or not.

Test Pattern	Attributes			Class
	Habits	Eats	Footwear	
1	Quiet	Baked	Clogs	Student
2	Quiet	Roasted	Sandals	Professor
3	Gabby	Roasted	Clogs	Professor
4	Quiet	Roasted	Clogs	Professor
5	Gabby	Fried	Clogs	Student

Please ensure that the names of all team members are on your submission! Please submit a word/PDF document with your responses to the above questions. In addition, please submit your .ARFF files for the training and test sets in question 2. Please submit your assignment to [LMS](#)